Portfolio Diversification using a Correlation Matrix

How low-correlation assets reduce portfolio risk · Microsoft Excel

Aditya Kushwaha

Tools	Microsoft Excel – CORREL, COVARIANCE.S, SUMPRODUCT, INDEX/MATCH, colour-scale heatmap, charts
Assets	Apple, Microsoft, JPMorgan Chase, Coca-Cola, ExxonMobil, gold (GLD)
Data	60 monthly total returns, Nov 2021 – Oct 2026
Outputs	Correlation heatmap, pair ranking, diversification benefit by portfolio, diversification curve

Monthly total returns (dividends reinvested) from chartrow.com monthly-returns tables. Correlations change over time; results are for learning, not investment advice.

1. Project overview

Objective. Measure how six assets move together and show, with numbers, how combining low-correlation assets lowers portfolio risk.

Correlation runs from –1 (opposite moves) to +1 (identical moves). When correlation is below +1, part of one asset's swings is offset by another's, so a portfolio is less volatile than the weighted average of its holdings.

Questions the project answers

- Which assets move together and which move independently?
- How much risk does diversification remove in different portfolios?
- How does portfolio risk change as lower-correlated assets are added?

Data and sources

The same 60 monthly returns as the Markowitz project. Statistics and covariance sheets provide the inputs for portfolio risk.

2. Steps taken – summary

#	Step	What was done
1	Collect returns	60 monthly total returns for six assets
2	Build the correlation matrix	CORREL for all 36 cells; heatmap from green (low) to red (high); average correlation per asset
3	Rank all pairs	15 pairs ordered from lowest to highest correlation using INDEX/MATCH into the matrix
4	Compare portfolios	Six portfolios from 100% Apple to a low-correlation mix; volatility vs weighted-average asset volatility
5	Draw the diversification curve	Add assets one by one in order of lowest correlation and track risk
6	Interpret	Link correlation levels to the size of the diversification benefit

Step 1 – Correlation matrix

Correlation Matrix of Monthly Returns

CORREL(asset i, asset j). +1 = move together perfectly, 0 = unrelated, –1 = move opposite.

	AAPL	MSFT	JPM	KO	XOM	GLD	Avg. correlation with others
AAPL	1.00	0.51	0.23	0.20	0.06	0.05	0.21
MSFT	0.51	1.00	0.29	0.07	(0.08)	0.11	0.18
JPM	0.23	0.29	1.00	0.18	0.24	(0.03)	0.18
KO	0.20	0.07	0.18	1.00	0.34	0.27	0.21
XOM	0.06	(0.08)	0.24	0.34	1.00	0.01	0.11
GLD	0.05	0.11	(0.03)	0.27	0.01	1.00	0.08
Average pairwise correlation (all 15 pairs)					0.16		

Reading the matrix

- *Apple and Microsoft (both large-cap technology) are the most correlated pair.*
- *Gold has the lowest average correlation (about 0.08) and is close to zero with AAPL, JPM and XOM – the strongest diversifier here.*
- *ExxonMobil moves almost independently of the technology stocks – energy follows oil prices, not tech earnings.*
- *Only two pairs are slightly negative (MSFT–XOM, JPM–GLD); 13 of 15 pairs are below 0.3, leaving plenty of room to diversify.*

Snapshot 1 – Correlation matrix heatmap with average correlation per asset

- Average pairwise correlation is only 0.16 – a well-diversified group.
- Apple–Microsoft (0.51) is the most correlated pair: both are large-cap technology stocks.
- Gold has the lowest average correlation (0.08); Microsoft–ExxonMobil (–0.08) and JPMorgan–gold (–0.03) are slightly negative.

Step 2 – Pair ranking

All Asset Pairs Ranked by Correlation

Pairs ordered from lowest to highest correlation; values are live INDEX lookups into the Correlation Matrix.

Rank	Asset 1	Asset 2	Correlation	Diversification value
1	MSFT	XOM	(0.08)	High
2	JPM	GLD	(0.03)	High
3	XOM	GLD	0.01	High
4	AAPL	GLD	0.05	High
5	AAPL	XOM	0.06	High
6	MSFT	KO	0.07	High
7	MSFT	GLD	0.11	Medium
8	JPM	KO	0.18	Medium
9	AAPL	KO	0.20	Medium
10	AAPL	JPM	0.23	Medium
11	JPM	XOM	0.24	Medium
12	KO	GLD	0.27	Medium
13	MSFT	JPM	0.29	Medium
14	KO	XOM	0.34	Low
15	AAPL	MSFT	0.51	Low

Snapshot 2 – All 15 asset pairs ranked by correlation

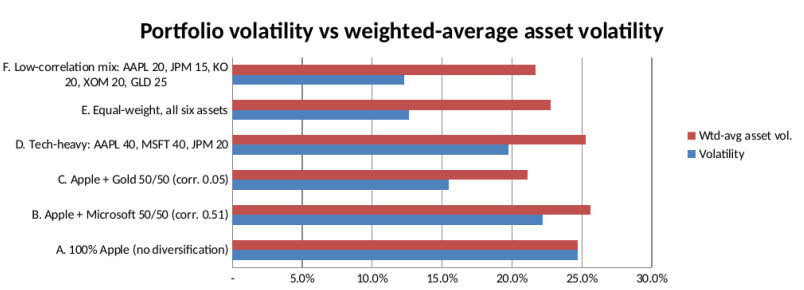
What this snapshot shows. Each correlation is a live INDEX/MATCH lookup into the matrix, and an IF formula labels the diversification value as High (below 0.1), Medium (0.1–0.3) or Low.

Step 3 – Diversification analysis

Diversification Analysis

Portfolio volatility = √(wᵀ Σ w). Diversification benefit = 1 – portfolio volatility ÷ weighted-average asset volatility.

Portfolio	AAPL	MSFT	JPM	KO	XOM	GLD	Total	Exp. return	Volatility	Sharpe	Wtd-avg asset vol.	Diversification benefit
A. 100% Apple (no diversification)	100.0%	-	-	-	-	-	100.0%	20.3%	24.7%	0.66	24.7%	-
B. Apple + Microsoft 50/50 (corr. 0.51)	50.0%	50.0%	-	-	-	-	100.0%	16.8%	22.2%	0.58	25.6%	13.2%
C. Apple + Gold 50/50 (corr. 0.05)	50.0%	-	-	-	-	50.0%	100.0%	18.5%	15.5%	0.94	21.1%	26.7%
D. Tech-heavy: AAPL 40, MSFT 40, JPM 20	40.0%	40.0%	20.0%	-	-	-	100.0%	17.3%	19.7%	0.67	25.3%	21.9%
E. Equal-weight, all six assets	16.7%	16.7%	16.7%	16.7%	16.7%	16.7%	100.0%	18.8%	12.6%	1.17	22.8%	44.5%
F. Low-correlation mix: AAPL 20, JPM 15, KO 20, XOM 20, GLD 25	20.0%	-	15.0%	20.0%	20.0%	25.0%	100.0%	19.7%	12.3%	1.28	21.7%	43.4%



Snapshot 3 – Portfolio risk vs weighted-average asset risk

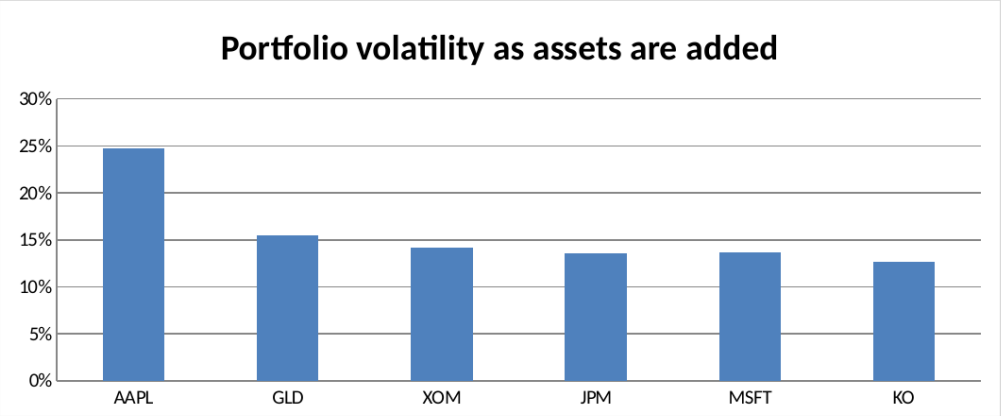
- 100% Apple: 24.7% volatility, no diversification benefit.
- Apple + Microsoft (correlation 0.51): risk falls only 13% to 22.2%.
- Apple + gold (correlation 0.05): risk falls 27% to 15.5% – the same two-asset split, but a much lower correlation.
- Equal-weight six assets: 12.6% volatility, a 44.5% reduction; the low-correlation mix reaches 12.3% with the best Sharpe ratio (1.28).

Step 4 – Diversification curve

Diversification Curve – Adding Lower-Correlated Assets

Equal-weight portfolios, adding assets in order of lowest average correlation with the assets already held.

No. of assets	Asset added	AAPL	MSFT	JPM	KO	XOM	GLD	Total	Exp. return	Volatility	Sharpe
1	AAPL	100.0%	-	-	-	-	-	100.0%	20.3%	24.7%	0.66
2	GLD	50.0%	-	-	-	-	50.0%	100.0%	18.5%	15.5%	0.94
3	XOM	33.3%	-	-	-	33.3%	33.3%	100.0%	21.9%	14.2%	1.26
4	JPM	25.0%	-	25.0%	-	25.0%	25.0%	100.0%	21.2%	13.6%	1.27
5	MSFT	20.0%	20.0%	20.0%	-	20.0%	20.0%	100.0%	19.6%	13.6%	1.14
6	KO	16.7%	16.7%	16.7%	16.7%	16.7%	16.7%	100.0%	18.8%	12.6%	1.17



Snapshot 4 – Portfolio volatility as lower-correlated assets are added

- Volatility drops from 24.7% (Apple only) to 15.5% with gold, 14.2% with ExxonMobil and 13.6% with JPMorgan.
- Adding Microsoft (highly correlated with Apple) does not reduce risk further (13.7%); adding Coca-Cola brings it to 12.6%.
- Most of the benefit comes from the first few low-correlation assets.

Why correlation matters
For two assets, $\sigma^2_p = w_1^2\sigma_1^2 + w_2^2\sigma_2^2 + 2w_1w_2\rho\sigma_1\sigma_2$. The lower the correlation ρ, the smaller the last term and the lower the portfolio risk.

Key findings

- Diversification depends on correlation, not just the number of holdings: two tech stocks diversify far less than a tech stock plus gold.
- An equal-weight portfolio of the six assets cut risk by about 45% compared with the average asset.
- Gold and energy were the best diversifiers for a technology-heavy portfolio over 2021–2026.
- Correlations can rise in market sell-offs, so diversification benefits should be monitored over time.

Conclusion

The correlation matrix is a simple but powerful tool: it shows which assets overlap and which add genuine diversification. Building portfolios from low-correlation assets reduced risk substantially without giving up return, demonstrating the core idea behind Modern Portfolio Theory.

Resume summary of this project

- Analysis of correlations between assets
- Practice in building a correlation matrix and a diversified portfolio
- Key learning: how low-correlation assets reduce portfolio risk

Skills shown

- Correlation and covariance
- Heatmaps with conditional formatting
- Portfolio volatility and diversification ratio
- INDEX/MATCH lookups and IF classification

Limitations

- Correlations are estimated from 60 months and change over time; no transaction costs or rebalancing effects.